Direct Corpus Interaction - Agentic Search without a Retriever

Seminar STOIS · WS 2026/27

M.Sc. Intelligent Systems · Ulm University of Applied Sciences

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- **Key Concepts:**
  - What are LLMs and context windows?
  - What is a Retriever-based agent?
  - What is a DCI agent?
- **Motivation:**
  - Large document collections can exceed an LLM's context window.
  - A retriever determines which search results reach the agent.
  - DCI lets the agent search and inspect raw documents directly using terminal tools.

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- **Defined Tasks:**
  - *How does Direct Corpus Interaction work?*
  - *How did DCI develop from coding agents and how has later research extended it?*
  - *Does DCI's reported advantage carry over to a self-hosted open-weight LLM?*
  - **Added Value:**
    - **Reproduction:**
      - **Agents :**
        - *Retriever-based*
          - Github Repo: /texttron/BrowseComp-Plus
          - BM25 or Qwen3- Embedding-8B
        - *DCI-based*
          - Github repo: /DCI-Agent/DCI-Agent-Lite
      - **Controlled comparison:**
        - Answering Model – t.b.d
        - Judging Model – t.b.d
        - BrowseComp-Plus with the full 100k-document corpus

Model	Accuracy	Δ Accuracy
Retrieval Agents		
GPT-o3	66.0	↓ 3.0
GPT-5.4 nano	44.9	↓ 24.1
GLM-4.7	50.2	↓ 18.8
MiniMax-M2.5	64.2	↓ 4.8
Qwen3.5-122B	58.7	↓ 10.3
Claude Haiku 4.5	45.8	↓ 23.2
Claude Sonnet 4.5	46.6	↓ 22.4
Claude Sonnet 4.6	69.0	–
DCI Agents		
DCI-Agent-Lite (GPT-5.4 nano)	62.9	-
DCI-Agent-CC (Claude Sonnet 4.6)	**80.0**	↑ 11.0

Figure 1: Reproduced from the DCI-Agent-Lite GitHub README, Table 1.

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- **Report:**
  - Accuracy
  - Trajectory metrics:
    - *Coverage and localization*
  - Token usage and number of turns to answer the question
- **Identify where and how often the local model fails**

Added Value: Challenging the topic further on Local-Models:

1. *Context-window size:*
   - How do context limits and context-management settings affect DCI accuracy on a local model?
   - Context-Management Settings:
     - *Truncation :*
       Does shortening search results remove useful evidence?
     - *Compaction:*
       Does clearing older tool output help the agent continue searching or remove information it still needs?
2. *Unrestricted Bash-Commands:*
   - Does the local model struggle with complex Bash-Commands?

Method	Avg. tools ↓	Cost / q ($) ↓	Acc. ↑
Retrieval Agents (GPT-5.4-nano)			
BM25	19	0.0527	32
Qwen3-Embedding-8B	**18**	0.0498	45
DCI-Agent-Lite (GPT-5.4-nano)			
`read + grep` (L4)	19	**0.0355**	61
Open bash (L4)	35	0.1021	**73**

Figure 2: Reproduced from Li et al. (2026), Table 5.